AccessLens v2: Welfare Access Risk Simulation, Eligibility Discovery, and Explainable AI Interface

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Abstract

This technical report presents AccessLens v2, a policy research system for simulating non-financial barriers in accessing government welfare schemes and discovering potential eligibility. The system combines a deterministic rule-based simulation model with counterfactual analysis to evaluate policy interventions. Unlike predictive AI systems, AccessLens v2 maintains strict separation between simulation and explanation: the core model is transparent and reproducible, while AI is used only for post-hoc explanation, translation, and natural language interaction. This document reports on the current implemented prototype, including structured datasets, access risk modeling, batch simulation capabilities, and visualization outputs. It also presents an expanded design for future development, incorporating an eligibility discovery module, expanded scheme coverage, policy lever simulation, multilingual AI explanation, and an interactive dashboard. The system prioritizes transparency, privacy preservation, and policy experimentation, serving as a decision-support tool for inclusive governance.

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1 Introduction

Government welfare schemes are designed to provide essential support to citizens across domains including health, housing, education, agriculture, and social protection. However, a persistent gap exists between eligibility and actual access. While financial eligibility criteria receive considerable policy attention, non-financial barriers often determine whether entitled individuals successfully obtain benefits.

These barriers manifest in multiple forms:

- **Awareness barriers**: Citizens may lack knowledge about scheme existence, purpose, eligibility criteria, or application procedures.
- **Documentation barriers**: Required documents such as income certificates, residence proofs, identity cards, or caste certificates may be difficult or costly to obtain.
- **Digital access barriers**: Online-only application systems exclude individuals without internet connectivity, digital devices, or necessary literacy skills.
- **Institutional dependency barriers**: Reliance on intermediaries, local officials, or bureaucratic processes can create friction and opportunities for exclusion.

Understanding how these barriers interact and affect different population segments is essential for policy makers, civil society organizations, and researchers seeking to improve scheme reach. Simulation-based systems offer a valuable methodology: they enable exploration of "what-if" scenarios without requiring real-world experimentation, identify leverage points for intervention, and support evidence-based policy discussions.

AccessLens v2 addresses this need through a transparent, reproducible framework for simulating access barriers, evaluating policy interventions, and communicating results through explainable AI interfaces. This report documents both the current implemented prototype and an expanded design for future development.

2 Motivation and Policy Context

The motivation for AccessLens v2 emerges from observed gaps in welfare scheme implementation. Studies across multiple contexts indicate that eligible individuals often fail to access benefits due to factors unrelated to their formal eligibility status. These access failures have significant consequences for welfare outcomes and perpetuate cycles of disadvantage.

Key observations motivating this work include:

- Access barriers are systematic and predictable, varying across population segments and scheme types.
- Policy interventions are often designed without clear understanding of which barriers matter most for which groups.
- Data on access barriers is scattered across sources and rarely integrated for systematic analysis.
- AI systems in governance contexts risk opacity and lack of accountability if not designed with transparency as a core principle.

AccessLens v2 responds to these challenges by providing a structured framework for barrier analysis, intervention simulation, and accountable AI explanation. The system is designed as a decision-support tool that informs rather than automates policy decisions.

3 System Design Philosophy

AccessLens v2 is constructed on foundational principles that distinguish it from conventional AI systems:

- **Deterministic modeling**: The core access risk model uses explicit rules and parameters rather than machine learning. This ensures reproducibility and complete explainability of system behavior.
- **Transparency and interpretability**: All model components, parameters, and assumptions are documented and accessible. Users can inspect why a particular access risk score or eligibility suggestion was generated.
- **Separation of simulation and explanation**: The simulation engine operates independently of AI. AI is used only after simulation to generate natural language explanations, never to determine outcomes or compute scores.
- **Privacy preservation**: The system uses synthetic personas rather than real individual data. No personal information is collected, stored, or processed. The eligibility discovery module suggests schemes based on user-provided attributes without storing them.
- **Counterfactual reasoning**: Policy analysis is conducted through systematic comparison of baseline scenarios with modified conditions, enabling exploration of intervention effectiveness.
- **Accessibility through multilingual support**: Explanations are provided in multiple Indian languages to ensure accessibility for diverse stakeholders.

These principles ensure that AccessLens v2 functions as a responsible AI system that supports human decision-making rather than replacing it.

4 Dataset Architecture

AccessLens v2 operates on three structured datasets that form the foundation for all simulations and analyses. Each dataset is designed to capture different aspects of the access ecosystem.

4.1 Schemes Dataset (schemes_v0.csv)

The schemes dataset contains structured information about welfare schemes collected from official government sources, scheme guidelines, and published documentation. Table 1 describes the fields in this dataset.

Table 1: Schemes Dataset Fields

Field	Type	Description
scheme_id	string	Unique identifier for each scheme
scheme_name	string	Official name of the scheme
domain	string	Policy area (health, education, housing, etc.)
application_mode	string	Online, offline, or hybrid
documents_required_count	integer	Number of documents typically required
requires_income_certificate	boolean	Whether income proof is required
requires_residence_proof	boolean	Whether residence proof is required
digital_only	boolean	Whether application is exclusively online
assisted_access_available	boolean	Whether help desks or assisted channels exist
steps_count	integer	Approximate number of application steps
source_type	string	Type of source document
source_confidence	string	High/medium/low confidence in data accuracy

Data collection prioritized publicly available information from official government portals and published scheme guidelines.

4.2 Barrier Taxonomy Dataset (barriers_v0.csv)

The barrier taxonomy dataset defines the types of access barriers that can impede scheme uptake. Table 2 describes the barrier categories and dataset fields.

Table 2: Barrier Taxonomy Dataset

Field	Type	Description
barrier_id	string	Unique identifier
barrier_type	string	awareness, documentation, digital, institutional
affected_stage	string	Stage in access process where barrier applies
default_severity	float	Baseline friction value (0-1 scale)
Barrier Categories:		
awareness		Lack of information about scheme existence or procedures
documentation		Difficulty obtaining required paperwork
digital		Technology-related access barriers
institutional		Reliance on intermediaries or bureaucratic processes

Each barrier type can be parameterized to reflect different levels of severity across contexts.

4.3 Synthetic Personas Dataset (personas_v0.csv)

The synthetic personas dataset contains constructed profiles representing different population segments. Using synthetic data ensures privacy while enabling meaningful simulation of how barriers affect different groups. Table 3 describes the persona attributes.

Table 3: Synthetic Personas Dataset

Field	Type	Description
persona_id	string	Unique identifier
literacy_level	string	Low, medium, high
digital_access	string	None, limited, full
document_completeness	float	Fraction of required documents readily available (0-1)
institutional_dependency	string	Low, medium, high reliance on intermediaries

Synthetic personas are designed to represent realistic variation in the population while avoiding any association with actual individuals.

5 Eligibility Discovery Module

The eligibility discovery module is a planned extension that will identify which welfare schemes an individual is likely eligible for based on provided attributes. This module operates on a set of user-input characteristics:

- Age
- Gender
- Income category (Below Poverty Line, Above Poverty Line, etc.)
- Occupation (farmer, laborer, self-employed, etc.)
- Rural/urban location
- Education level
- Farmer status (landholding size, crop type)
- Disability status
- Household size
- Social category (if relevant for scheme targeting)

The module scans scheme eligibility criteria encoded in the expanded schemes dataset and produces a ranked list of potentially applicable schemes. Important design principles include:

- **Suggestion, not determination**: The system explicitly disclaims official eligibility decisions. Outputs are suggestions requiring verification.
- **Privacy preservation**: User attributes are not stored; matching occurs in memory and results are presented without retention.
- **Transparency**: Each suggestion includes the matching criteria that triggered it, enabling users to understand why a scheme was suggested.
- **Coverage indication**: Where scheme criteria are ambiguous or incomplete, confidence levels indicate suggestion reliability.

The eligibility module integrates with the access risk model to provide not only scheme suggestions but also assessments of access difficulty for each suggested scheme.

6 Access Risk Model

The core of AccessLens v2 is a deterministic model that computes access friction scores for each combination of scheme and persona. The model operates through three mechanisms:

- **Scheme barrier activation**: Each scheme specifies which barriers are relevant to its access process, encoded through scheme attributes.
- **Persona amplification**: Persona attributes determine how strongly each barrier affects that individual.
- **Barrier severity values**: Each barrier has a base severity that can be modified by counterfactual conditions.

The access risk score is calculated as:

$$\text{AccessRisk} = \frac{\sum_{i=1}^{n} \left(\text{Severity}_i \times \text{Activation}_{i,\text{scheme}} \times \text{Amplification}_{i,\text{persona}}\right)}{\sum_{i=1}^{n} \text{Severity}_i}$$

Where:

- n is the number of barrier types
- Severity_i is the baseline severity of barrier i (0-1 scale)
- $\text{Activation}_{i,\text{scheme}}$ is 1 if barrier i applies to the scheme, 0 otherwise
- $\text{Amplification}_{i,\text{persona}}$ captures how persona attributes affect susceptibility to barrier i (0-1 scale)

The normalized score ranges from 0 to 1, with higher values indicating greater access difficulty. This formulation ensures interpretability while maintaining deterministic behavior.

7 Counterfactual Simulation Engine

The counterfactual simulation engine enables systematic evaluation of policy interventions by comparing baseline access risk scores with modified scenarios. Each counterfactual represents a specific policy change:

- **CF1 — Awareness Improvement**: Reduces severity of awareness barriers by a configurable factor, simulating the effect of information campaigns or outreach programs.
- **CF2 — Documentation Simplification**: Reduces documentation barriers by modifying the amplification factor for document completeness, representing policy changes that streamline paperwork requirements.
- **CF3 — Assisted Access Removal**: Increases barriers for schemes that previously had assisted access by removing the mitigating effect of assisted channels, modeling what happens when support systems are withdrawn.
- **CF4 — Digital Enablement**: Reduces digital barriers by modifying amplification factors for digital access, simulating improved internet connectivity, digital literacy programs, or device availability.

For each counterfactual, the system recalculates access risk scores with modified parameters. The difference between counterfactual and baseline scores (delta) indicates the potential impact of the intervention:

$$\Delta_{\text{CF}} = \text{AccessRisk}_{\text{CF}} - \text{AccessRisk}_{\text{Baseline}}$$

Negative delta values indicate reduced access difficulty (improvement), while positive values indicate increased difficulty.

8 Batch Simulation Pipeline

The batch simulation pipeline executes the access risk model across all combinations of schemes and personas in the datasets. For each combination, the pipeline:

1. Retrieves scheme attributes and barrier activation patterns
2. Retrieves persona attributes and amplification factors
3. Computes baseline access risk score
4. Applies each counterfactual scenario (CF1-CF4)
5. Computes counterfactual scores and deltas
6. Exports results to structured CSV format

Batch outputs include:

- Scheme and persona identifiers
- Baseline access risk scores
- Counterfactual scores for each intervention
- Delta values for each intervention
- Aggregated statistics across schemes and personas

This systematic approach enables comprehensive analysis of access patterns and intervention effectiveness across the entire design space.

9 Visualization and Analytical Insights

Preliminary visualizations generated from batch simulation outputs reveal patterns and insights:

9.1 Persona vs Scheme Risk Heatmap

The heatmap visualization displays access risk scores across all scheme-persona combinations, revealing concentration of access difficulty among vulnerable groups. Darker regions indicate higher risk, enabling identification of which persona types face greatest barriers for which scheme categories.

9.2 Average Risk by Persona

Aggregating across schemes shows which persona attributes correlate with highest access friction. Preliminary analysis indicates that digital access and document completeness are strong predictors of access difficulty, with personas lacking these facing consistently higher risk scores.

9.3 Average Risk by Scheme

Scheme-level aggregation identifies characteristics associated with high barriers. Digital-only schemes with multiple documentation requirements show consistently elevated risk scores, suggesting these design features contribute significantly to access difficulty.

9.4 Counterfactual Effectiveness Ranking

Comparison of average delta values across interventions reveals which policy changes have largest potential impact. Early analysis suggests:

- Digital enablement (CF4) shows greatest improvement for personas with limited digital access
- Documentation simplification (CF2) benefits multiple persona types across scheme categories
- Assisted access removal (CF3) significantly increases risk for personas with high institutional dependency

This last finding is particularly significant: it suggests that support systems play a critical role in enabling access for vulnerable populations, and their removal could disproportionately harm those most in need.

10 AI Explanation and Translation Layer

The AI explanation layer is a planned extension that will operate exclusively on structured simulation outputs. Its purpose is to make simulation results accessible to diverse stakeholders through natural language and multilingual support.

10.1 Explanation Generation

Given simulation outputs for a specific scheme-persona combination, the AI will generate explanations addressing:

- Overall access risk level and interpretation
- Dominant barriers contributing to risk
- How persona attributes affect each barrier
- Counterfactual outcomes and their implications
- Comparative context relative to similar schemes or personas

Example explanation: "This persona faces high access difficulty (risk score 0.78) primarily due to digital barriers, as the scheme requires online application and the persona has limited internet access. Documentation requirements also contribute moderately. If digital access were improved (CF4), risk would decrease by 0.3, suggesting this intervention would be most effective."

10.2 Multilingual Translation

All explanations will be generated in multiple Indian languages to ensure accessibility for stakeholders including:

- Hindi
- Marathi
- Tamil
- Bengali
- Telugu
- Kannada

Translation will preserve technical accuracy while adapting to linguistic and cultural contexts.

10.3 Design Principles

- **Post-hoc only**: AI receives only simulation outputs, never raw data or model parameters
- **Verifiable**: Explanations can be traced back to underlying simulation numbers
- **Model-agnostic**: Different AI models can be used interchangeably
- **Conservative**: AI avoids adding information beyond simulation outputs

11 Natural Language Query System

The natural language query system will enable users to interact with AccessLens v2 through plain language questions. The system will:

1. Parse user queries to extract intent and parameters
2. Map queries to appropriate system functions (eligibility lookup, risk simulation, counterfactual analysis)
3. Execute relevant simulations or retrievals
4. Generate natural language responses with optional visualization
5. Support follow-up questions and refinement

Example queries:

- "Which schemes am I eligible for if I am a farmer in rural Maharashtra with low income?"
- "What barriers make it hardest for elderly applicants to access health schemes?"
- "If we simplify documentation requirements, how much does access improve for Below Poverty Line households?"
- "Show me schemes with high access barriers in urban areas"

The query system will maintain the separation between interpretation and computation: query parsing uses AI, but all numerical results come from the deterministic simulation engine.

12 System Architecture

Figure 1 illustrates the overall system architecture of AccessLens v2, showing the flow from data through simulation to user interaction.

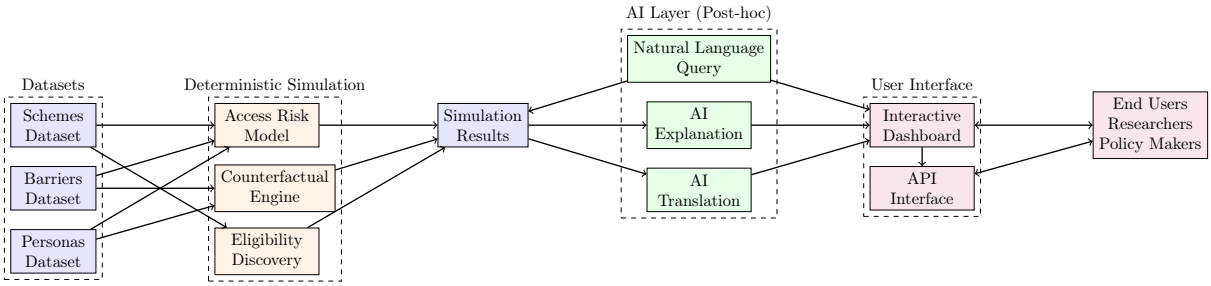

Figure 1: AccessLens v2 System Architecture

The architecture maintains clear separation between deterministic simulation components and AI explanation layers, ensuring transparency and reproducibility of core analytical functions.

13 Ethical Considerations

AccessLens v2 is designed with ethical considerations as foundational requirements:

- **No personal data**: The system uses only synthetic personas for simulation. The eligibility discovery module accepts user-provided attributes but does not store them.
- **No eligibility decisions**: The system suggests potentially eligible schemes but explicitly disclaims official eligibility determination. All outputs require verification through official channels.
- **Limited AI role**: AI is used only for explanation and translation after simulation completes, never for decision-making or score computation.
- **Transparent assumptions**: All model parameters and assumptions are documented, enabling scrutiny and debate.
- **Advisory outputs**: System results are intended to inform policy discussions and individual exploration, not to automate or replace human judgment.
- **Accessibility**: Multilingual support ensures that language barriers do not prevent stakeholders from engaging with system outputs.
- **Accountability**: Clear separation between components enables auditing and verification of system behavior.

These safeguards ensure that AccessLens v2 contributes to responsible AI practice in governance contexts while avoiding common pitfalls of automated decision systems.

14 Limitations

AccessLens v2, in its current and planned form, has several limitations that should be acknowledged:

- **Scheme coverage**: The dataset includes a limited number of schemes; findings may not generalize to all welfare programs.

- **Simplified barrier model**: The deterministic model captures major barrier types but may omit nuanced or context-specific factors.
- **Synthetic personas**: While privacy-preserving, synthetic personas may not capture the full complexity of real population diversity.
- **No field validation**: Simulation results have not been validated against real-world access patterns.
- **Static data**: Scheme information may become outdated as policies change.
- **AI limitations**: AI explanations may contain inaccuracies or oversimplifications despite post-hoc design.
- **No causal claims**: Simulations show associations and potential effects but cannot establish causality.

These limitations define the scope of appropriate system use and highlight directions for future improvement.

15 Future Work

Several extensions are planned for subsequent development phases:

1. **Expanded scheme dataset**: Increase coverage to 100+ schemes across multiple states and domains, with regular updates.
2. **Enhanced persona modeling**: Develop more sophisticated persona attributes capturing intersectional vulnerabilities.
3. **Additional counterfactuals**: Implement more nuanced policy interventions including phased implementation and combination scenarios.
4. **Full AI integration**: Complete implementation of explanation, translation, and natural language query capabilities.
5. **Interactive dashboard**: Develop Streamlit-based web interface with visualizations, filters, and export capabilities.
6. **API development**: Create REST API for programmatic access to simulation capabilities.
7. **Field validation**: Collaborate with civil society organizations to compare simulation results with observed access patterns.
8. **Longitudinal tracking**: Incorporate scheme update mechanisms to maintain current information.
9. **User studies**: Evaluate system usefulness with policy makers, researchers, and potential beneficiaries.
10. **Open source release**: Package system for public use with documentation and examples.

These developments will transform AccessLens v2 from a research prototype into a practical tool for policy analysis and inclusive governance.